A Multi-Agent System for Time-Series Modelling across Scientific and Financial Domains

Mid-semester deliverable: the agent framework, and its first capability validated end to end

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Abstract. The goal of this project is an agent that performs time-series modelling across unrelated domains, choosing the right analysis for a question instead of running a fixed pipeline. This semester I built the framework and validated it on the first of its tasks. The framework is a registry of 20 models spanning 10 task types and 4 domains, with a capability filter that selects among them. Across 123 labelled queries that filter routes correctly 95.9% of the time against 83.7% for an LLM, in under a second against 41 minutes; combining the two gives 95.1%, which is worse than the filter alone. I then used the framework to take one task, anomaly detection, all the way through on two domains. These are 440 Kepler light curves labelled by NASA ephemerides and seven years of Indian equity returns labelled by a pre-registered event calendar, evaluated under held-out splits. Every detector falls within 0.04 PR-AUC of a rule that learns nothing, and a supervised model given the labels does no better; the gap between in-sample and held-out scores rises monotonically with model capacity, from 1.0× to 6.4×. The limitation is representational rather than algorithmic. A Box Least Squares period search recovers 61% of orbital periods across all 440 targets at median significance 42.

1. Project Goal and Objectives

Time-series questions come in several kinds. *What will this be worth next month?* is forecasting. *Which days were unusual?* is anomaly detection. *How likely is an event in the next 24 hours?* is a point-process problem. *How much could I lose?* is risk. Each needs a different model, and the right model also depends on the domain the data came from.

Most systems handle this with a fixed pipeline per domain. The goal here is different: **one agent that decides, for any question, which analysis to run, and chooses on measured evidence rather than a language model's judgement.** The registry currently spans 10 task types across astronomy, finance and solar physics. This report delivers that framework and takes one task, anomaly detection, through full validation on two domains.

Objectives. O1. Build a multi-agent system that routes a natural-language question to the correct analysis with no language model in the decision path. **O2.** Assemble anomaly detection datasets for astronomy and finance whose labels do not depend on the series being scored. **O3.** Measure a broad set of detectors on both domains under a protocol strict enough that the result can be trusted. **O4.** Identify what actually limits performance, and demonstrate a method that overcomes it.

Scope of this report. The agent framework and anomaly detection on astronomy and finance. Solar-flare forecasting, the risk layer and the cross-sectional equity work are under way but belong to the second half, and appear only in the plan in Section 7.

2. Literature Review

Benchmarks in this field are known to be unreliable. Wu and Keogh [1] examined the Yahoo, Numenta, NASA and OMNI archives and found most problems suffer from at least one of four flaws: triviality, unrealistic anomaly density, mislabelled ground truth, and run-to-failure bias. Roughly 90% yield to a single line of arithmetic, and their conclusion is that on such data no level of reported performance implies a method is useful.

The scoring protocol is also compromised. Kim et al. [2] showed that point adjustment, the dominant convention, lets a random anomaly score reach state-of-the-art F1, because marking a whole segment correct for one lucky hit rewards noise.

Complex models do not reliably win. A 2024 benchmark of 1,070 series across 40 datasets [3] concluded that simpler architectures and statistical methods often outperform neural ones. For transit detection specifically, Box Least Squares [4] remains the standard method and the matrix profile [5] is the canonical tool for subsequence anomalies. I use both, and the contrast between them turns out to be informative.

Gap addressed here. These critiques are widely cited but rarely applied by authors to their own data. I run the four-flaw audit on my own datasets before reporting any result, and report in-sample and held-out scores side by side so that the difference between them is visible.

3. System Design and Methodology

A question is parsed into a *domain* and a *task*; the registry is filtered to models that can actually run given the available inputs; one is selected on recorded skill; it executes; and a rule-based explanation is composed. Only the final phrasing step calls a language model. Each of the 21 registry entries declares its domain, task, required inputs, return type and measured skill, so adding a model is one registry entry rather than an edit to a branch.

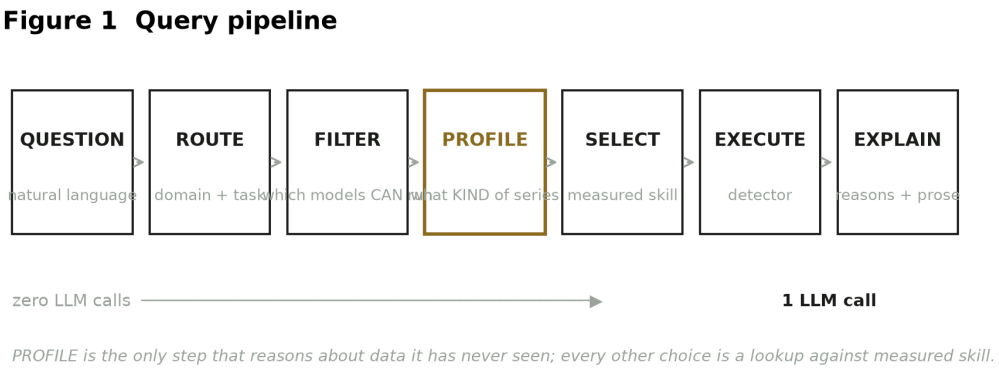


Figure 1. Every decision except the final phrasing is made in Python. Profiling is the only step that reasons about data the system has never seen.

Profiling. Before choosing a detector the agent inspects the series itself. The Lomb–Scargle peak-to-background ratio separates repeating from one-off structure cleanly: over ten Kepler targets and ten equities the two populations sat at medians of 1,939 and 12.4 with no overlap, and a threshold classified all twenty correctly. This matters because the decision inverts the right detector. Scored as a discord, the matrix profile reaches ROC 0.303 on KIC 10074466, below chance; scored as a *motif*, 0.697. A transit recurs 221 times and is therefore the *least* unusual feature of its own light curve.

Routing accuracy. The capability filter reaches 95.9%; the LLM alone 83.7%; the two combined 95.1%. The filter is not merely cheaper, it is more accurate, and adding the language model costs accuracy as well as time.

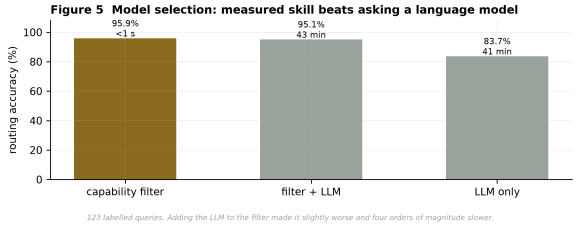


Figure 2. Model selection on 123 labelled queries.

3.3 Datasets, labelling and protocol

Data and labels. Astronomy: 440 Kepler light curves (217 confirmed planets, 223 false positives), windowed at 12 hours with a 3-hour stride, giving 384,281 windows at 24.4% positive. A window is labelled a transit if it overlaps T■ + nP from NASA's published ephemeris, orbital mechanics, never the brightness being scored. Finance: an equal-weighted index of the 50 most liquid NSE names, 2019–2026, windowed at 21 trading days, 1,891 windows at 2.2% positive, labelled from a pre-registered calendar of budget sessions, election results and ten documented shocks.

Avoiding circular labels. An earlier version labelled astronomy windows by thresholding their own *depth*, then evaluated detectors that search for deep dips; measured inflation on discovery was +0.11 AUC. The finance equivalent is labelling a day anomalous because its return was large. Solar flares are excluded for the same reason in its strongest form: a flare's class is *defined* by an X-ray threshold, so detecting flares from X-ray flux cannot fail.

Audit and protocol. I audited both datasets against Wu and Keogh's four flaws [1] *before* any result was recorded. Equity passes all four; Kepler warns only on anomaly density, since short-period planets transit almost continuously. PR-AUC is my decision metric because it is floored at the base rate, with ROC-AUC reported beside it, and I do not apply point adjustment: Kim et al. [2] showed that protocol lets a random score reach state-of-the-art F■. Splits are temporal for equity and by-target for Kepler, stratified by per-target prevalence. An unstratified split produced halves at 17.9% and 42.8% positive, which makes their PR-AUC values incomparable.

4. Implementation

The system is written in Python and organised so the domain-neutral components cannot see domain-specific information. The `anomaly` package holds the scoring contract and windowing, 22 detectors across six families, the label sources, the four-flaw audit, held-out splitting with prevalence stratification, the Lomb-Scargle structure profiler, and a parallelised grid runner. `astronomy/transit_search` implements the Box Least Squares period search, and `agents/model_registry` holds 21 models with declared capabilities and measured skill.

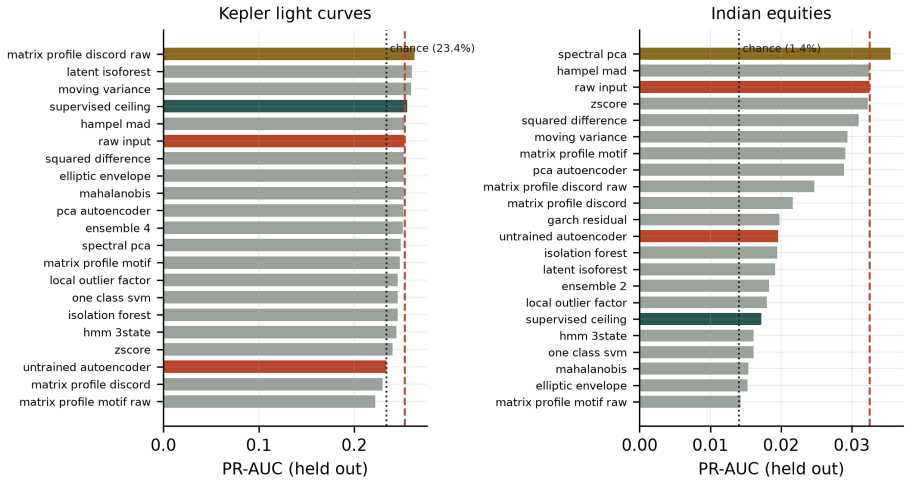
Detectors span thresholds (z-score, Hampel/MAD), density (isolation forest, Mahalanobis, local outlier factor, one-class SVM, elliptic envelope), reconstruction (PCA and LSTM autoencoders, spectral PCA, latent isolation forest), latent-state (hidden Markov model, GARCH residuals) and four matrix profile configurations, plus two deliberate no-skill floors and a supervised ceiling. Results are served through a Streamlit dashboard and a chatbot, and all 440 astronomy targets are precomputed into MongoDB, giving 4.5 ms lookups at query time.

5. Results and Discussion

Almost nothing clears the floor. On Kepler the best detector beats the best no-skill floor by 0.0102 PR-AUC, and the supervised ceiling, a gradient-boosted model trained on the labels, reaches only 0.2555 and is outscored by three unsupervised detectors. On

equity the margin is 0.0030.

Figure 2 Detector comparison, held-out



red dashed = best no-skill floor; dotted = chance. On Kepler every detector lies within 0.04 of a rule that learned nothing.

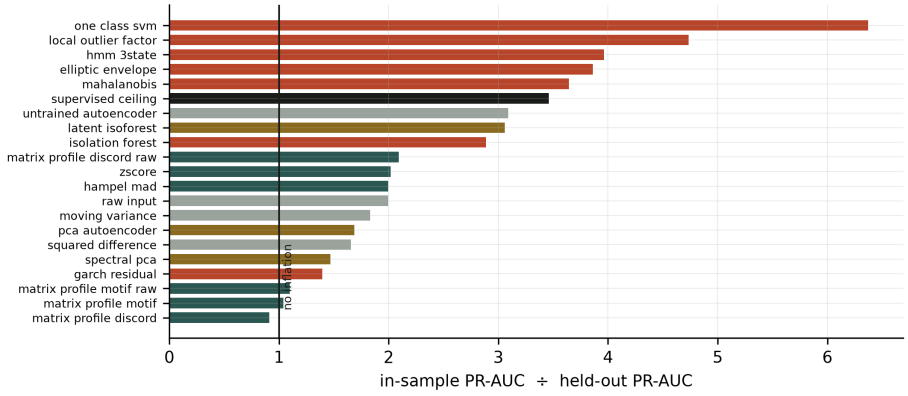
Figure 3. Twenty-two detectors across five families, scored on held-out data.

Domain	Detector	PR-AUC	Lift	Inflation
Kepler	matrix profile discord raw	0.2630	1.12x	1.1x
Kepler	latent isoforest	0.2605	1.11x	1.2x
Kepler	moving variance	0.2595	1.11x	1.2x
Kepler	supervised ceiling (ceiling)	0.2555	1.09x	1.3x
Kepler	raw input (floor)	0.2528	1.08x	1.2x
Equity	spectral pca	0.0355	2.52x	1.5x
Equity	raw input (floor)	0.0325	2.31x	2.0x
Equity	hampel mad	0.0325	2.31x	2.0x
Equity	local outlier factor	0.0180	1.28x	4.7x
Equity	one class svm	0.0161	1.14x	6.4x

Table 1. Held-out PR-AUC for selected detectors. Lift is PR-AUC divided by the base rate; inflation is in-sample PR-AUC divided by held-out.

In-sample scores lie in proportion to capacity. The ratio of in-sample to held-out PR-AUC rises monotonically with how much a model can memorise: 1.0–1.1× for the parameter-free matrix profile, 2.0× for threshold rules, 2.9× for isolation forest, 4.7× for local outlier factor and 6.4× for a one-class SVM. Evaluated transductively, which is the standard convention in unsupervised anomaly detection, the one-class SVM appears to be the best equity detector at 4.62× lift. Held out, it manages 1.14×. The single apparent win for sophistication was an artefact of capacity.

Figure 3 How much each detector's in-sample score overstates it



Inflation rises with model capacity: parameter-free methods sit near 1.0, a one-class SVM at 6.4x.

Figure 4. In-sample scores overstate held-out performance in proportion to model capacity.

Sequence models add nothing. Every detector above treats a window as an unordered summary. An LSTM sequence-to-sequence autoencoder was added specifically to test whether temporal order inside a window carries signal. It reached 1.92× on equity, below the linear autoencoder's 2.06× and a threshold rule's 2.31×, at roughly 250× the compute.

The representation is the limit. On Kepler, 44.2% of labelled transits are no more extreme than a typical non-transit window at this resolution, and the supervised ceiling reaches only 1.09× chance. Changing the *question* rather than the algorithm resolves it. Box Least Squares folds the light curve on thousands of candidate periods and stacks every repetition. Across all 440 targets it yields a significant periodogram peak for 91.4% and the published period to within 1% for 60.9%, at median significance 42.3 and median period error 0.005%. The gap comes from harmonic locking. Around 33.6% return P/2, P/3 or 2P, so the signal is present and only the fold multiple is wrong.

Figure 4 Why a period search succeeds where anomaly detection fails

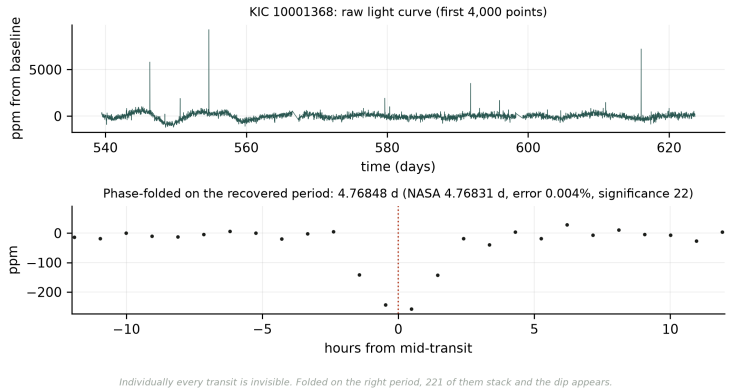


Figure 5. A single transit is invisible in the raw series. Folded on the recovered period, 221 of them stack into a clear 250 ppm dip.

Configuration inverts between domains. The same algorithm needs opposite settings in the two domains. On Kepler the matrix profile must be read as a *motif* under z-normalised distance, because transits repeat and their depth varies by star. On equity it must be read as a *discord* under un-normalised distance, because shocks are unique and the size of the move is the signal. Getting both axes wrong scores 0.95×, below chance. This is the clearest evidence that the right detector is a property of the data's structure rather than of the scientific domain it came from.

6. Conclusion

Three things hold up. A capability filter over measured skill routes more accurately than a language model while costing nothing, and adding the model degrades it. Under held-out evaluation with non-circular labels, twenty-two detectors across two domains sit within 0.04 PR-AUC of a rule that learns nothing, their in-sample advantages scaling with capacity rather than skill. And the binding constraint is representational: a matched-filter period search recovers 61% of orbital periods where no generic detector clears 1.12× chance.

7. Work Plan for the Second Half

The second half fills in the remaining tasks, taking each to the same evidential standard as anomaly detection. The agent gains a forward-looking task: risk assessment conditioned on a Markov regime model, where a regime-conditional Value-at-Risk passes Kupiec and Christoffersen coverage tests that the unconditional version fails. It also gains a third domain, blazar light curves from the Fermi-LAT repository, which the pipeline accepts without code changes and which my structure profiler predicts should behave like equities rather than like Kepler.

References

[1] R. Wu and E. J. Keogh. Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress. *IEEE TKDE*, 2022.

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[5] C.-C. M. Yeh et al. Matrix profile I: all pairs similarity joins for time series. *ICDM*, 2016.